



SOFTWARE REQUIREMENTS SPECIFICATION & UI/UX DESIGN SYSTEM SPECIFICATION

Product: PropKart Real Estate Operating System & Desk CRM
Release: Version 2.1.1 (Production Build 10+)

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Parent Organization	NB Property Tech
Target Audience	Lead Graphics Designer, Principal UI/UX Designer, Frontend Engineers
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Document Classification	Internal / Proprietary — Product & UI/UX Baseline
Target Environment	Production Release v2.1.1 (Flutter Material 3 / CRM Design System)

1. EXECUTIVE SUMMARY & SYSTEM ARCHITECTURE

1.1 Product Philosophy & Value Proposition

PropKart is not a public classifieds portal. It is an enterprise-grade, high-performance Real Estate Operating System & Desk CRM built specifically for property brokers, sales teams, telecallers, and agency owners.

The platform provides a unified workspace to manage both Rental and Re-Sale inventory at scale. It handles the complete deal lifecycle: multi-channel ad lead ingestion (Meta Lead Ads, Google Sheets), intelligent demand matchmaking, one-tap WhatsApp sharing, site visit scheduling, negotiation management, and deal closing documentation.

1.2 Core Product Pillars of v2.1.1

- **Unified Dual Inventory:** Seamless management across residential and commercial books with dedicated visual atmospheres.
- **Automated Matchmaking:** Real-time pairing of buyer/tenant requirements with active inventory using budget brackets (10% tolerance), BHK configurations, furnishing, and localities.
- **One-Tap Client Sharing:** Instant generation of branded multi-property PDF dossiers, WhatsApp broadcast messages, and trackable public web shortlists.
- **Direct Marketing Ingestion:** Webhooks for Meta Lead Ads and Google Sheets with lead intelligence, duplicate phone detection, and 1-click promotion to CRM.
- **Role-Aware Security & Permission Matrix:** 4 operational roles governed by a dynamic 40-metric permission control center.
- **Internal Team Messenger:** Dedicated real-time communication desk (/messages) with role filtering and Super Admin thread oversight.

1.3 Technical Architecture Overview

- **Front-End Framework:** Flutter (Dart ^3.11.1) with Material 3 and custom CRM Design System.
- **State Management:** BLoC (flutter_bloc 9.1.1) for domain events + Riverpod for reactive caching.
- **Persistence:** Isar embedded NoSQL database for rapid offline querying + flutter_secure_storage for encrypted JWT credentials.
- **Target Platforms:** Web (PWA at propkart.nbpropertytech.com), Android, iOS, Windows Desktop, macOS Desktop.

2. DESIGN SYSTEM & UI TOKEN ARCHITECTURE

2.1 Dual-Theme Architecture & Color Palette

PropKart implements runtime theming via ThemeManager, supporting Light and Dark modes across two master presets with an Atmosphere Shift between Rental and Re-Sale desks.

Token Role	Modern Teal (Default SaaS)	PropKart Classic (Terracotta)	Dark Mode Override
primary	#159B73 (Emerald Teal)	#C15D4A (Terracotta Clay)	#10B981 / #D47A66
primaryHover	#128764	#A64C3C	#34D399 / #E08B78
accent	#E8F5F1 (Mint Wash)	#F6D8D0 (Sand Tint)	#0D3D31 / #3A2824
rentalAtmosphere	#159B73 (Fresh Teal)	#5F8064 (Sage Green)	#10B981 / #5F8064
resaleAtmosphere	#C15D4A (Warm Terracotta)	#8B4513 (Saddle Wood)	#D47A66 / #8B4513
canvas	#F8FAFC (Slate Canvas)	#F4F4F3 (Warm Alabaster)	#0F172A (Deep Slate)
surface	#FFFFFF (Pure White)	#FFFFFF (Neutral Card)	#1E293B (Midnight Surface)
border / divider	#E8ECF2 (Light Gray)	#E3DDD7 (Muted Sand)	#334155 (Slate Border)
textPrimary	#14213D (Deep Navy)	#1A1A1A (Carbon Charcoal)	#F8FAFC (Off-White)
textSecondary	#64748B (Slate Muted)	#68738A (Clay Charcoal)	#94A3B8 (Cool Slate)
whatsappGreen	#25D366 (Official)	#25D366 (Official)	#25D366
success / danger	#10B981 / #E11D48	#5F8064 / #DC2626	#10B981 / #E11D48

UX Directive — Atmosphere Shift: When the desk operates in **Rental Mode**, primary buttons, status pills, and active filter tabs accent in Teal / Sage. When toggled to **Re-Sale Mode**, the accent shifts to Terracotta / Clay. This prevents agents from accidentally pitching rental properties to resale buyers.

Token	Size	Weight	Line Ht	Tracking	Usage Context
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largeDisplay	32 pt	Bold (700)	40 pt	-0.02 em	Splash, Hero KPIs, Auth welcome
pageTitle	24 pt	Bold (700)	32 pt	-0.015 em	Screen headers (Properties, Leads)
sectionTitle	18 pt	SemiBold (600)	24 pt	-0.01 em	Card headers, modal titles
cardTitle	15 pt	SemiBold (600)	20 pt	-0.01 em	Property cards, requirement codes
bodyMedium	14 pt	Medium (500)	20 pt	0.0 em	Inputs, data table cell text
body	13.5 pt	Regular (400)	18 pt	0.0 em	Descriptions, remarks, chat
captionBold	12 pt	SemiBold (600)	16 pt	+0.01 em	Status badges, table headers
caption	11 pt	Regular (400)	14 pt	+0.01 em	Timestamps, subtitles
footnote	10 pt	Medium (500)	12 pt	+0.02 em	Unread counts, validation text

2.2 Spacing, Radii, Shadows & Breakpoints

- **Spacing Scale:** xs: 4px | s: 8px | m: 12px | l: 16px | xl: 20px | xxl: 24px | xxxl: 32px | huge: 48px | mega: 64px.
- **Corner Radii:** xs: 4px | s: 8px (Inputs) | m: 12px (Cards) | l: 16px (Panels) | xl: 20px (Modals) | pill: 999px (Chips).
- **Elevation & Shadows:** soft: 0px 2px 8px (0.04) | medium: 0px 6px 16px (0.08) | floating: 0px 12px 32px (0.12) | primaryGlow: 0px 4px 14px (0.35).
- **Glassmorphism:** Navigation & overlays utilize BackdropFilter (sigma 20) with 82% surface opacity fill.
- **Breakpoints:** Mobile: < 768px | Tablet: 768px – 1023px | Desktop: 1024px – 1439px | Ultrawide: >= 1440px (1440px max width).

3. GLOBAL APPLICATION SHELL ARCHITECTURE

3.1 Resizable Sidebar & Glass Top Bar

- **Resizable Sidebar (ModernSidebar):** 245px default width (resizable to 320px via vertical drag handle; collapses to 70px icon rail). Houses logo brand lockup, 9 core module links with dynamic unread badge counters, and UserProfileCard.
- **Glass Modern Top Bar (ModernTopBar):** 74px desktop height with 420px debounced omni-search field, Quick Add (+) circular button with glow, Notifications center drawer, Team Messages shortcut (/messages), and Super Admin profile dropdown.
- **Mobile Bottom Navigation Bar:** On viewports < 768px, persistent 5-tab bottom navigation bar: Dashboard, Properties, Quick Add (+), Leads, Profile.

4. ROLE-BASED ACCESS CONTROL (RBAC) & PERMISSION MATRIX

PropKart v2.1.1 features an enterprise-grade Permission Matrix Service (PermissionMatrixService) with 40+ granular controllable permissions across 7 categories.

Feature / Capability	Super Admin	Admin	Sales	Telecaller
Dashboard, Properties, Leads	YES	YES	YES	YES
Rent vs Re-Sale Switch	YES	YES	YES	YES
Create & Edit Properties	YES	YES	YES	YES
Soft Delete to Recycle Bin	YES	YES	NO	NO
Restore / Permanent Drop	YES (Full)	YES (Restore)	NO	NO
Verify Property (Blue Check)	YES	YES	NO	NO
Export to Excel (.xlsx)	YES	YES	NO	NO
Unmasked Client Phone	YES	YES	YES	Masked Option
Campaign Inbound Inbox	YES	YES	NO	YES (Intake)
Manage Webhook Secrets	YES	YES	NO	NO
1-Click Move to CRM	YES	YES	NO	YES
Executive Reports & BI	YES	YES	NO	NO
Live Permission Matrix	YES (Exclusive)	NO	NO	NO

5. COMPLETE SCREEN-BY-SCREEN FUNCTIONAL SPECIFICATIONS

5.1 Authentication & Gatekeeper Screens

- **/splash:** High-resolution brand mark with pulsating shimmer animation (CRMMotion.nameShimmer). Executes startup Isar sync warmup and validates 9-hour session inactivity timeout.
- **/get-started:** Apple-style presentation card with 'Sign In to Desk' primary CTA, Terms and Privacy footer.
- **/login:** Email with regex validation, password visibility toggle, rate-limiting banners, and forgot password link.
- **/reset-password:** Consumes recovery query tokens (token_hash, code) with live password strength meter.
- **/terms-and-conditions & /privacy-policy:** Typographic document readers rendering embedded legal markdown.

5.2 Executive Dashboard (/dashboard)

- **Welcome Header:** Time-of-day greeting (Good Morning / Afternoon / Evening) + user full name + formatted date + Rent vs Re-Sale atmosphere switch.
- **KPI Metric Cards:** 4-column desktop / 2-column mobile grid: Total Listed Inventory (+% growth indicator), Available Units (% of stock), Deals Closed (rented/sold), Active Requirements, Registered Team Roster count.
- **Today's Schedule Card:** Client appointments agenda with time, property code link, client name, assigned executive, status pill, and WhatsApp action.
- **Priority Follow-ups Card:** Sub-tabs (Today, Due, Future) with client budget, configuration, and direct phone/WhatsApp shortcuts.
- **Analytics Section:** Residential vs Commercial inventory mix pie chart + Top Localities ranked bar chart.
- **Desk Quick Notes:** Scratchpad checklist with optimistic check/uncheck updates for agent reminders.

5.3 Properties Inventory Hub (/properties & /properties/:id)

- **Atmosphere & Category Strip:** Rent vs Re-Sale toggle, Residential vs Commercial tabs, BHK chips (1 BHK, 2 BHK, 3 BHK, 4+ BHK, RK, Plot, Office, Shop).
- **Desk Toolstrip:** Omni-search, Locality multi-select, Price range sliders (■ Lacs / ■ Crores), Image status filter (With Images, No Images), 'My Added Only' toggle, Card Grid vs Data Table switcher, Compare checkboxes.
- **Property Card Anatomy:** Thumbnail carousel with counter (1/8), formatted price header, size badge (Super Built-up & Carpet), Verified Blue Badge, property code, owner WhatsApp contact row, action menu (Edit, Share, Delete).
- **Side-by-Side Comparison Modal:** Evaluates up to 4 properties across 15 structured parameters (Price, Deposit, Carpet, Floor, Age, Facing, Furnishing, Parking, Bathrooms, Balconies, Maintenance).
- **Add / Edit Property (AddEditPropertyScreen):** 7-section form: 1. Identity, 2. Location & Map Pin Picker (lat/long reverse geocoding), 3. Dimensions, 4. Commercials (1-6 month deposit calculator), 5. Specs & Amenities, 6. Media (compression & Cloudinary upload), 7. Ownership & Remarks. Integrated CRMDraftRepository for autosave restoration.
- **Property Detail Screen (/properties/:id):** Full-bleed gallery, spec chips, map view, owner quick-action card, client sharing drawer.

5.4 Recycle Bin (/bin)

- **Soft Delete Vault:** Displays deleted property listings and client requirements with deletion timestamp and retention countdown.
- **Actions:** 1-click Optimistic Restore back to active inventory or Permanent Drop (Super Admin exclusive).

5.5 Leads & Requirements Hub (/requirements)

- **Main Tabs:** Active Leads, Requirements, Scheduled Follow-ups.
- **14-Stage Deal Progression Funnel:** Tracks leads through 14 standardized states: 1. Lead Created -> 2. Requirement Added -> 3. Requirement Verified -> 4. Matching Started -> 5. Properties Matched -> 6. Properties Shared -> 7. Client Viewed -> 8. Client Interested -> 9. Site Visit Scheduled -> 10. Site Visit Completed -> 11. Negotiation -> 12. Booking Token -> 13. Documentation -> 14. Closed / Won.
- **Automated Matchmaking Engine:** Evaluates client budget (10% tolerance), localities, BHK, furnishing, and facing against active inventory. Computes Match Readiness Score (Ready, Needs Information, Cannot Match) and Requirement Quality Metric (High, Medium, Low, Poor).
- **Client Sharing Suite:** Branded PDF Dossier Generator (PropertySharePdf), Public Web Shortlist Link (/share/:sessionId) with view tracker, and pre-templated WhatsApp broadcast.
- **Follow-up System:** Calendar & clock alarm picker modal. Strictly isolates external followups from internal_crm_remarks audit trail.

5.6 Campaign & Inbound Marketing Ingestion (/campaign)

- **Dual-Section Isolation:** Segregates inbound records into two non-overlapping pipelines: Property Listing Leads vs Requirement Leads.

- **Lead Intelligence Engine:** Parses raw text responses from Meta custom questions and Google Forms to extract budget, BHK, and localities. Detects duplicate phone numbers across the database.
- **Interactive Status Dropdown:** Follow-up (opens schedule modal), Interested (green highlight & 1-click Move to CRM), Not Interested (moves to archive tab).
- **1-Click 'Move to CRM':** Promotes campaign lead into a permanent property or requirement while preserving ad attribution metadata (meta_lead_id, campaign_name).
- **Connections Screen (/campaign/connections):** Meta Lead Ads webhook setup, Verify Token, Google Sheets sync URL, and JSON payload simulator.

5.7 Internal Team Messenger (/messages)

- **Layout:** Master-detail responsive split layout (Roster list 320px on left, chat thread on right).
- **Super Admin Team Grouping:** Roster grouped by teams with 'Propkart Admin' pinned first. Super Admins can inspect cross-team threads.
- **Roster Filtering:** All, Admins, Telecallers, Sales with unread message badges and active status indicators.
- **Chat Canvas:** Branded chat bubbles, timestamps, delivery checkmarks, and keyboard shortcuts (Enter to send, Shift+Enter for newline).

5.8 Reports & Business Intelligence (/reports)

- **Global Filter Bar:** Date presets (Today, Yesterday, Last 7 Days, This Month, Custom Range, All Time) + Team/Agent dimension filter.
- **Overall Business Insight (/reports/leads/overall-business-insight):** Executive revenue projections, 14-stage lead conversion funnel, callback velocity metrics, team leaderboard & rankings, and lead source attribution donut chart.
- **Export Options:** 1-click export of executive dossiers in .xlsx and formatted .pdf.

5.9 Shared Document Libraries (/library)

- **Rental Library (/rental-library):** Lease agreements, tenant/landlord KYC proofs, deposit receipts, utility clearances, inspection media.
- **Re-Sale Library (/resale-library):** Sale deeds, title clearances, society NOCs, floor plans, tax receipts, loan sanction letters.
- **Service Agent Library (/service-agent-library):** Contractor agreements, GST certificates, SLAs, and completed work showcases for electricians, plumbers, and cleaning crews.

5.10 Directories & Team Administration

- **Client Index (/clients):** Buyer/tenant directory with Kanban Pipeline vs Data Table switcher.
- **Property Owners (/owners):** Landlord registry linked to properties owned with rental yield summaries.
- **Builders & Developers (/builders):** Developer directory with tier ratings, active projects, and POCs.
- **Employees & Users (/users):** Staff directory, user creation modal, password reset queue tab, and agent activity drawer.
- **Settings & Governance (/settings):** Profile editor, Theme customizer with 'Make as System Default', Location manager (cities/localities), Super Admin Permission Matrix Card, Security Audit Logs (/settings/audit-logs), and Sync Diagnostics (/settings/sync-debug).

6. UI/UX DESIGNER IMPLEMENTATION DIRECTIVES

- **Auto-Layout Mandatory:** Design responsive auto-layout frames in Figma across 390px (Mobile), 834px (Tablet), and 1440px (Desktop). Avoid fixed-width canvases.
- **Information Density:** Prioritize high data density, clear typography contrast, and compact padding. Avoid oversized empty hero sections.
- **Atmosphere Cues:** Distinct visual accents for Rental Mode (cool teal) vs Re-Sale Mode (warm terracotta).
- **Mandatory 5 States per Screen:** 1. Default Loaded, 2. Skeleton Loading (shimmering layout placeholders), 3. Empty State (friendly graphic + CTA), 4. Error State (icon + retry button), 5. Permission Denied (shield icon + admin contact).

7. MASTER ROUTE & PERMISSION INDEX TABLE

Screen Name	Route URL	Allowed Roles	Primary Layout
Splash & Sync	/splash	Public / System	Responsive Fullscreen
Get Started	/get-started	Unauthenticated	Centered Card
Login Screen	/login	Unauthenticated	Centered Auth Card
Dashboard	/dashboard	All 4 Roles	4-Col Grid / Mobile Stack
Properties CRM	/properties	All 4 Roles	Data Table & 3-Col Cards
Property Detail	/properties/:id	All 4 Roles	Hero Gallery & Split Specs
Leads & Matching	/requirements	All 4 Roles	Pipeline Cards & Table
Public Share Link	/share/:sessionId	Public Client	Responsive Web Portal
Campaign Inbox	/campaign/leads	SuperAdmin, Admin, TC	High-Density Table
Campaign Connectors	/campaign/connections	SuperAdmin, Admin	Form & Payload Inspector
Team Messenger	/messages	All 4 Roles	Master-Detail Split
Overall Business Report	/reports/leads/overall..	SuperAdmin, Admin	Charts & KPI Dossier
Document Library	/library	All 4 Roles	Category 3-Card Grid
Employees Directory	/users	SuperAdmin, Admin	User Cards & Roster
Recycle Bin	/bin	SuperAdmin, Admin	High-Density Table
System Settings	/settings	All 4 Roles	Multi-Tab Master Panel

8. DOCUMENT APPROVAL & SIGN-OFF

This document represents the formal software requirements specification and UI/UX handoff baseline for PropKart v2.1.1.

- **Product Architecture Approval:** Lead Systems Architect, NB Property Tech
- **UI/UX Design Lead Approval:** Principal UX Designer, NB Property Tech
- **Engineering Review:** Lead Flutter Engineer & Core API Team